

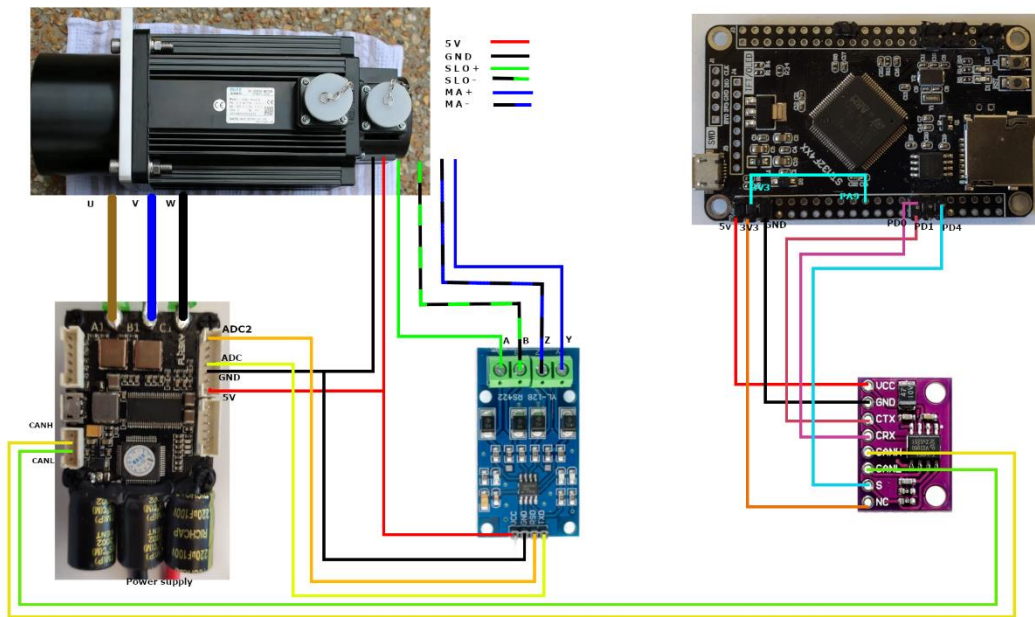
This is a tutorial step by step how to setup vesc with vesctool.

First, is to make all connections, wiring motor and encoder. NEVER TURN ON A VESC WITHOUT A MOTOR WIRED.

This is all connections that you need to make this work properly, from vesc and stm32.

Remember to use a brake circuit if you are not using batteries.

Is not mandatory to connect stm and tja1051 till OpenFFBoard setup.



Vesc setup

1-. After powering vesc and connect USB, open Vesc Tool. First step is to upload firmware. To

do that go to connection tab, select vesc COM and click connect  on right side. Now your vesc need to be connected.

2-. Time to upload. Go to Firmware tab, and first one we need to update is bootloader, to avoid vesc in limited mode. So, Firmware/Bootloader and select 412. Now press button

“Update firmware on the connected vesc” . It takes few seconds. Time to reboot, Terminal/Reboot (upper side of window). Every single time on this tuto you read reboot, need

to do this step, and of course connect again vesc .

Now is time to upload custom firmware, into Custom file tab, select your firmware (default, 005 or 0005 if you change shunt resistors) and again press “Update firmware on the connected vesc”. Reboot.

3-. In Motor Settings/General/General. Setup you motor type in FOC, select the encoder and resolution (22 for Biss-C).

4-. Motor Settings/General/current. select your motor current max (1A = 1.42NM, for example: 130ST-M15015 30Nm peak, so $30/1.42 = 21.13$. Use 20 -20 settings), and battery current max (10 -10 A 500W 48v PSU.)

5-. Motor Settings/General/Advanced. minimum input voltage 43 and maximum 50 (prevention if Brake Resistor fail and you are using previous PSU).

6-. Write motor configuration  (Right bar on screen). Reboot.

7-. Motor Settings/FOC/General. Sensor mode select encoder. Replace Alpha value, 30,

instead of 2000 ERPM/s. 

Now you can calibrate your motor. First RL, then Lambda and then apply.



8-. Motor Settings/FOC/ Encoder. Detect encoder  and then apply .

9-. Write motor configuration . Reboot.

10-. Motor Settings/General/Advanced. Set min current 0A. (with default shunt, cogging could appear). There is a Bug on vesc that sometimes refuse to save settings when this value is

" 0 ". Write motor configuration . Reboot.

If vesc don't save your settings after a reboot or power-cycle, change the min current to 0.5, make your setting update, write motor configuration , reboot, and change again this settings.

11-. Data Analysis/Realtime Data/Rotor position. When you select encoder , and turn your motor shaft, you need to see movement in the graph.

12-. In the bottom of the vesc tool : click on D 0.20 and play button



to run motor in dutycyle : it must turn in CW direction, click on



stop button to stop it . If not reverse the motor "general/motor" tab, and redo a encoder calibration.

13-. If you are here and the motor turn in dutycyle, you can now test the torque force. With

my mige 15015, I can run motor with $I = 1A$  and feel the force when you stop the shaft. Force is linear during a turn. If you feel force growth/reduce, you have a foc calibration issue.

Setup the OpenFFBoard to drive the motor

In the Configurator, choose the VESC as driver.

By default the CANBus speed is set to the default vesc CAN Bus so you don't need to change it. Check if the vesc status is ok and the encoder rate is around 1000, if not you can increase the canbus speed to increase it.

When you move the wheel, the bargraph have to move.

If you need, you can center it in the Axis tab: the VescDriver store the center position to keep it. If you want to erase the stored position, you have a button to reset it.